

- 5 (a) Fig. 5.1 is a photograph of a section through a kidney.



Fig. 5.1

On Fig 5.1, use label lines and letters to label the regions of the kidney where the following are located:

L – loop of Henlé

D – distal convoluted tubule

P – podocyte cells.

[3]

- (b)** In a cold environment, people urinate more frequently. This condition is called cold diuresis.

In a cold environment, homeostatic mechanisms keep heat loss to a minimum. Blood is diverted away from the skin so that more blood circulates around the main body organs, including the kidney.

Suggest **and** explain why cold diuresis occurs.

[4]

- (c) The water potential of the blood is maintained at a constant level by a mechanism called negative feedback.

With reference to the maintenance of the water potential of the blood, explain what is meant by negative feedback.

[5]

- (d) ADH affects the number of water channel proteins, aquaporins, in the cell surface membranes of the cells of the distal convoluted tubule.

State what happens to the water potential **and** volume of the urine if cells in the distal convoluted tubule have more aquaporins in their cell surface membranes.

water potential

volume of urine

[1]

[Total: 13]